

capable of destroying process-driven electronic modules in any system by generating resets or inducing power latchups, thereby neutralising targeted devices.

Another possible military application of DS-110 system includes disruption of electronic fusing mechanism of projectiles, artillery shells and IEDs. The basic system is omni-directional but it can be configured to provide directional coverage. The system packaged in a suitcase weighs 28 kilograms. It is powered by a built-in battery capable of powering the system for 30 minutes on a continuous basis or three hours in controlled bursts.

CHAMP. CHAMP, an abbreviation for Counter-electronics High-powered microwave Advanced Missile Project, is a technology demonstration programme of the US Air Force Research Laboratory (AFRL) jointly with Boeing Phantom Works for the development and field testing of counter electronics HPM aerial demonstrator, intended for integrating HPM payloads on airborne platforms such as cruise missiles and unmanned combat aerial vehicles.

CHAMP represents a class of a directed-energy HPM weapon in which the weapon travels towards the target rather than the target travelling towards the HPM weapon. Boeing is the prime contractor, airborne platform provider and system integrator. The HPM source and the pulsed-power system, respectively, come from Ktech Corp. of Albuquerque (directed-energy capability of Ktech Corp. is now acquired by Raytheon) and Sandia National Laboratories under a separate contract with AFRL.

CHAMP brings together directed-energy technology capabilities of AFRL and Boeing's missile design capabilities for development of a new breed of non-lethal but highly-effective weapon systems.

Military use of kinetic energy weapons to counter adversary's

command, control, communications and computer assets have an associated collateral damage. Collateral damage to civilian electronic equipment could be huge, depending on the location of the targeted electronic system, targeted spectrum bands and employment techniques. CHAMP that makes use of HPMs is a non-lethal alternative to conventional kinetic weapons to disrupt and destroy electronics assets.

While current kinetic energy weapons cannot penetrate hidden, underground targets, EMPs from an HPM weapon can penetrate through metal elements leading into underground command centres, to damage and even destroy sensitive components in command and control systems, communication systems and computers associated with the targeted systems. CHAMP provides the war-fighter a non-lethal, low collateral damage capability that can be used against targets that are currently on the kinetic restricted target list.

HPM payload of CHAMP has been successfully tested including ground testing and flight demonstrations. During the flight testing of CHAMP, the missile was pointed at a set of simulated targets. The test confirmed that the missile could be controlled and timed while using an HPM system against multiple targets and locations. Fig. 11 illustrates the concept of an HPM attack by CHAMP mounted on an aerial platform such as X-45 UAV.

In future, development efforts of CHAMP programme would lead to HPM payloads on cruise missiles or loitering platforms. These payloads would be capable of delivering multiple HPM bursts to a series of targets for applications such as suppression of enemy air defences, disruption and destruction of command and control networks and national infrastructure with minimised collateral damage.

To be continued...



SINCE 1962

INFRARED THERMOMETERS



IRT1050P

- Temp. Range: -50°C ~ 1050°C
- Accuracy: $\pm 1.5^\circ\text{C}$
- Distance Spot Ratio: 50 : 1
- Emissivity: 0.10 ~ 1.00 (Adjustable)

IRT550P

- Temp. Range: -50°C ~ 550°C
- Accuracy: $\pm 1.5^\circ\text{C}$
- Distance Spot Ratio: 12 : 1
- Emissivity: 0.95

IRT380P

- Temp. Range: -50°C ~ 380°C
- Accuracy: $\pm 1.5^\circ\text{C}$
- Distance Spot Ratio: 12 : 1
- Emissivity: 0.95

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